

Home Work 04: Linear Rational Quotient Derivative

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \frac{2x}{x+1}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — LINEAR RATIONAL FORMULATION

The problem presents a rational polynomial quotient with a linear monomial numerator and affine denominator:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = 2x, \quad v(x) = x + 1$$

Applying the fundamental Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(2x) = 2, \quad \frac{d}{dx}(x+1) = 1$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	$2x$	$\frac{d}{dx}(2x) = 2$
হর (v)	$x + 1$	$\frac{d}{dx}(x + 1) = 1$
হরের বর্গ (v^2)	$(x + 1)^2$	$(x + 1)^2$ (উৎপাদক আকারেই রাখা সমীচীন)

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. Linear Subtraction Simplification: Expand carefully:

$$2(x + 1) - 2x = 2x + 2 - 2x = +2$$

Avoid dropping the constant term 2 to mistakenly write 0 or $2x$.

2. Factored Form: Do not expand $(x + 1)^2$ into $x^2 + 2x + 1$. Leaving it in factored form is canonical in AMIE engineering exams.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{2x}{x+1} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \frac{(x+1) \frac{d}{dx}(2x) - (2x) \frac{d}{dx}(x+1)}{(x+1)^2}$$

ধাপ ৩: প্রমিত অন্তরজের মান $\frac{d}{dx}(2x) = 2$ এবং $\frac{d}{dx}(x+1) = 1$ বসিয়ে:

$$\frac{dy}{dx} = \frac{(x+1) \cdot 2 - (2x) \cdot 1}{(x+1)^2}$$

ধাপ ৪: লবের বন্ধনী বিস্তার ও পদসমূহ সরলীকরণ:

$$\frac{dy}{dx} = \frac{2x + 2 - 2x}{(x + 1)^2} = \frac{2}{(x + 1)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = \frac{2}{(x + 1)^2}$$

ENGLISH NOTE — DOMAIN, MONOTONICITY & ASYMPTOTES

- Domain of Function & Derivative:** Non-zero denominator requires $x + 1 \neq 0 \implies x \neq -1$. Thus, $\text{Dom}(y) = \text{Dom}\left(\frac{dy}{dx}\right) = \mathbb{R} \setminus \{-1\}$.
- Strict Increasing Monotonicity:** For all $x \neq -1$, $(x + 1)^2 > 0$, hence $\frac{dy}{dx} = \frac{2}{(x+1)^2} > 0$. The function is strictly increasing on its domain.
- Asymptotes:** Vertical asymptote at $x = -1$; Horizontal asymptote at $y = \lim_{x \rightarrow \pm\infty} \frac{2x}{x+1} = 2$.

ENGLISH NOTE — DETERMINANT & SYNTHETIC SHIFT DOUBLE CHECK

- Determinant Method:** $y = \frac{2x+0}{1x+1} \implies ad - bc = (2)(1) - (0)(1) = 2 \implies \frac{dy}{dx} = \frac{2}{(x+1)^2}$
- Synthetic Shift:** $y = \frac{2(x+1)-2}{x+1} = 2 - 2(x + 1)^{-1} \implies \frac{dy}{dx} = 0 - 2(-1)(x + 1)^{-2} = \frac{2}{(x+1)^2}$

AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Linear Differentiation	Accurate derivatives $\frac{d}{dx}(2x) = 2, \frac{d}{dx}(x + 1) = 1$	1.5
Algebraic Simplification	Simplifying $2x + 2 - 2x = 2$	1.0
Final Standard Format	Writing $\frac{2}{(x+1)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0